

Software Development Lifecycles

Planning stage

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Why we need SDLC ?

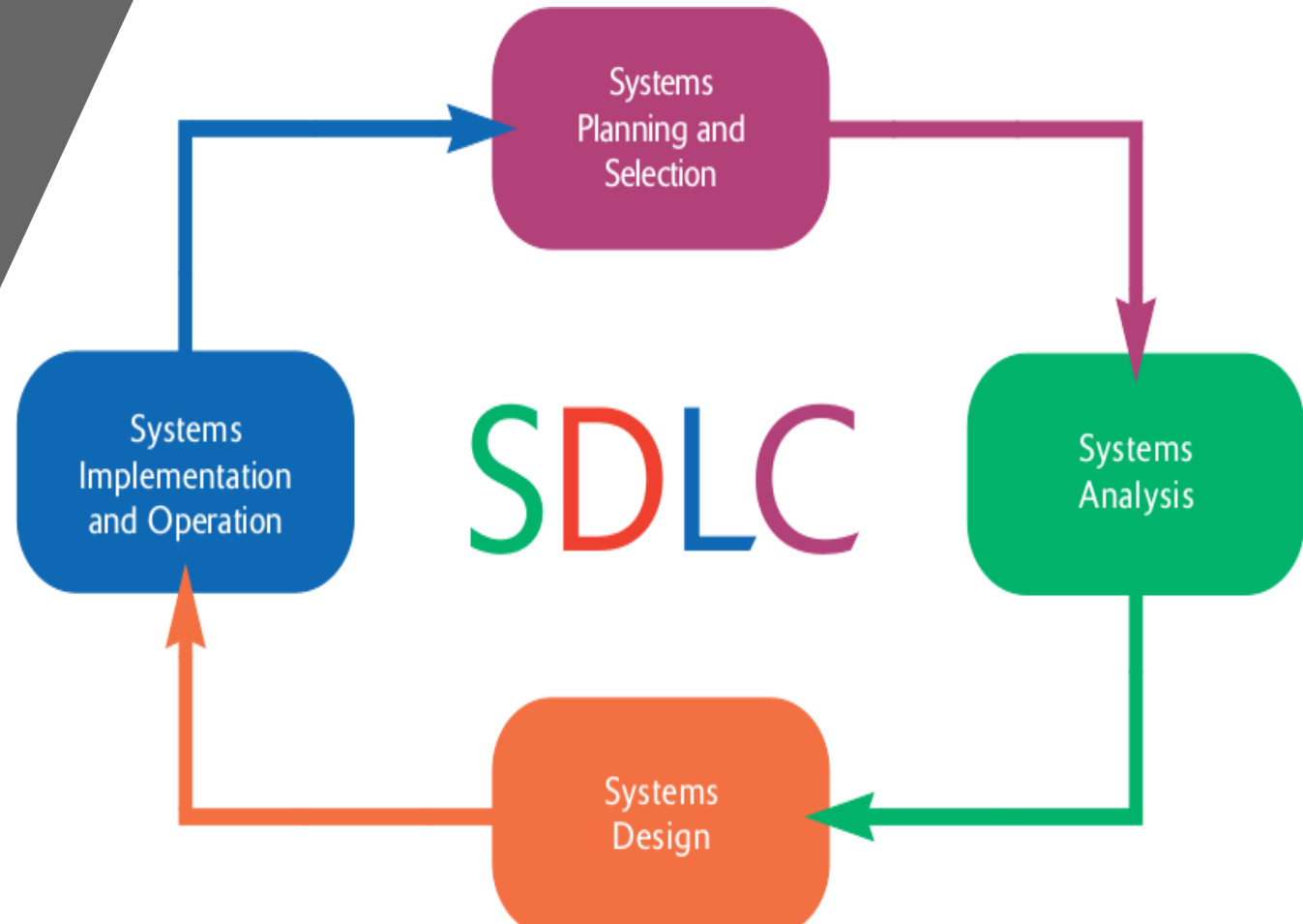
1. For better software quality,
2. Make the goals of the software clear to the team (developers know what to build and testers know what and how to test)
3. Reduce the rate of vulnerabilities (fewer or none)
4. Management control.
5. Effective documentation and reduced dependencies
6. Effective resource utilization.
7. Effective cost and time definition.
8. Ensure the architecture and design are secure.
9. Meet and exceed Customer's expectations.

Software Development life cycle (SDLC)

- **SDLC is a process used by the software industry to design, develop and test the quality of software.**
- **The SDLC aims to produce a high-quality software that meets or exceeds customer expectations, reaches completion within times and cost estimates.**
- **The SDLC defines a methodology for improving the quality of software and the overall development process.**

SDLC Stages

Four stages SDLC



SDLC Stages

Five stages SDLC



SDLC Stages

six stages SDLC



SDLC stages

Seven stages of SDLC

PLANNING

ANALYSIS

DESIGN

DEVELOPMENT

TESTING

DEPLOYMENT

MAINTENANCE



Planning

Project Planning

- ✧ The **project plan**, which is created at the start of a project, is used to **communicate how the work will be done to the project team and customers**, and to help assess progress on the project
- ✧ The objective of software project planning is to provide a framework that enables the manager to make **reasonable estimates of resources, cost, and schedule**.
- ✧ Estimates should attempt **to define best-case and worst-case scenarios** so that project outcomes can be bounded.
- ✧ The **plan must be adapted and updated as the project proceeds**.

Project planning (cont'd)

- Is a formal document designed to **guide the control and execution of a project**. A project plan is **the key to a successful project and is the most important document that needs to be created when starting any business project**.

At this stage we do the following:

- Determine the main purpose and goals of the project based on stockholders (software scope).
- Cost Estimation (feasibility study).
- Resource Allocator (stuffing, software and hardware).
- Project scheduling considering dependencies between tasks.
- Risk assessment and risk mitigation strategies.

Planning stages

✧ Planning is done:

- **At the proposal stage**, when you are bidding for a contract to develop or provide a software system. **The aim of planning at this stage is to provide information that will be used in setting a price for the system to customers.**
- **During the project startup phase**, when you have to plan **who will work on the project, how the project will be broken down into increments, how resources will be allocated across your company, etc.**
- **Periodically throughout the project**, when you **modify your plan in the light of experience gained and information from monitoring the progress of the work.**



Software Scope

- **Software scope** describes the **functions and features** that are to be delivered to end users.
- The **data that are input and output** “content” that is presented to users as a consequence of using the software.
- The **performance, constraints, interfaces, and reliability** that bound the system.
- **Dependency of the system on other systems** at the deployment environment.

Project scheduling

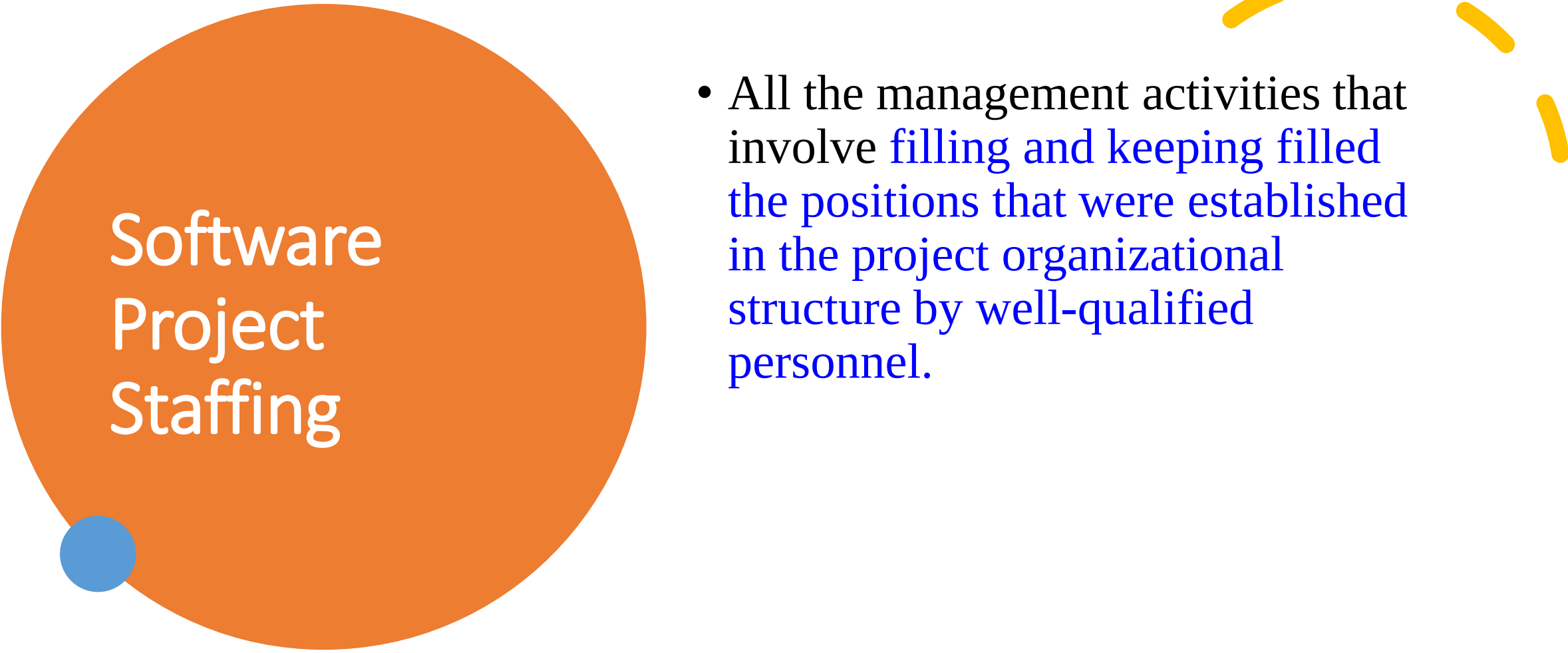
- ✧ **Project scheduling** is the process of deciding how the work in a project will be organized as **separate tasks**, and when and how these tasks will be executed.
- ✧ You estimate the calendar **time needed** to complete each task, the effort required, and **who will work** on the tasks that have been identified.
- ✧ You also have to estimate the **resources** needed to complete each task, such as the disk space required on a server, the time required on specialized hardware, such as a simulator, and what the travel budget will be.

Project scheduling activities

- ✧ Split project into tasks and estimate time and resources required to complete each task.
- ✧ Organize tasks concurrently to make optimal use of workforce.
- ✧ Minimize task dependencies to avoid delays caused by one task waiting for another to complete.
- ✧ Dependent on project managers intuition and experience

Milestones and deliverables

- ✧ **Milestones** are points in the schedule against which you can assess progress, for example, the handover of the system for testing.
- ✧ **Deliverables** are work products that are delivered to the customer, e.g. a requirements document for the system.



Software Project Staffing

- All the management activities that involve filling and keeping filled the positions that were established in the project organizational structure by well-qualified personnel.

Major Issues in Staffing

1. Project managers are frequently selected for their ability to program or perform engineering tasks rather than their ability to manage
2. The productivity of programmers, analysts, and software engineers varies greatly from individual to individuals.
3. Universities are not producing a sufficient number of computer science graduates who understand the software engineering process or project management.
4. Training plans for individual software developers are not developed or maintained

Factors to Consider when Staffing

- **Education:** Does the candidate have the minimum level of education for the job? Does the candidate have the proper education for future growth in the company?
- **Experience:** Does the candidate have an acceptable level of experience? Is it the right type and variety of experience?
- **Training:** Is the candidate trained in the language, methodology, and equipment to be used, and the application area of the software system?
- **Motivation:** Is the candidate motivated to do the job, work for the project, work for the company, and take on the assignment?

Factors to Consider when Staffing (cont'd)

- **Commitment:** Will the candidate demonstrate loyalty to the project, to the company, and to the decisions made?
- **Self-motivation:** Is the candidate a self-starter, willing to carry a task through to the end without excessive direction?
- **Group affinity:** Does the candidate fit in with the current staff? Are there potential conflicts that need to be resolved?
- **Intelligence:** Does the candidate have the capability to learn, to take difficult assignments, and adapt to changing environments?

Sources of Qualified Project staff



Transferring personnel from within the project itself, from task to another task



Transfers from other projects within the organization, when another project has ended or is being cancelled.



New hires from other companies through such methods as job fairs, referrals.

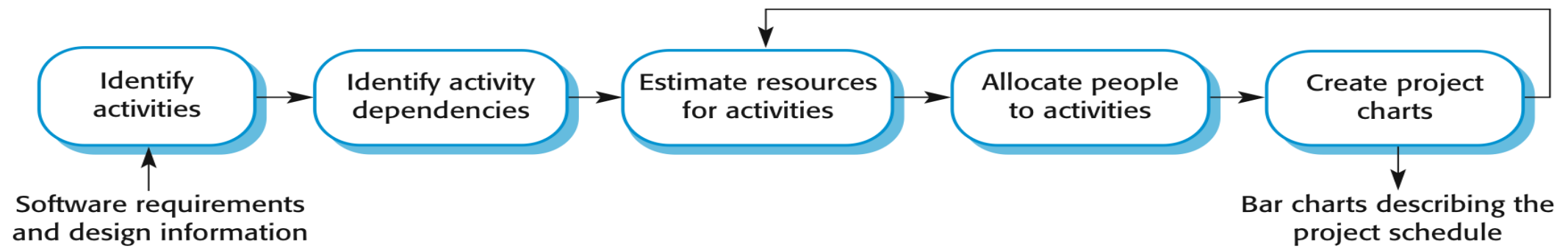


New college graduates can be recruited either through interviews on campus or through referrals from recent graduates who are now company employees.



If the project manager is unable to obtain qualified individuals to fill positions, one option is to hire unqualified but motivated individuals and train them for those vacancies

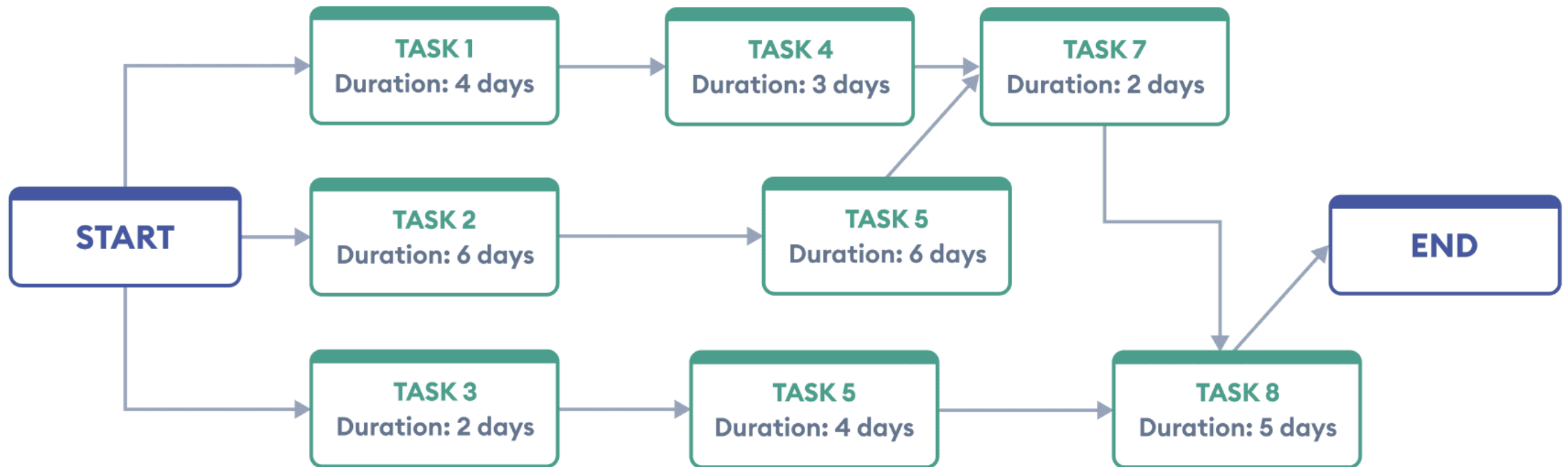
The project scheduling process



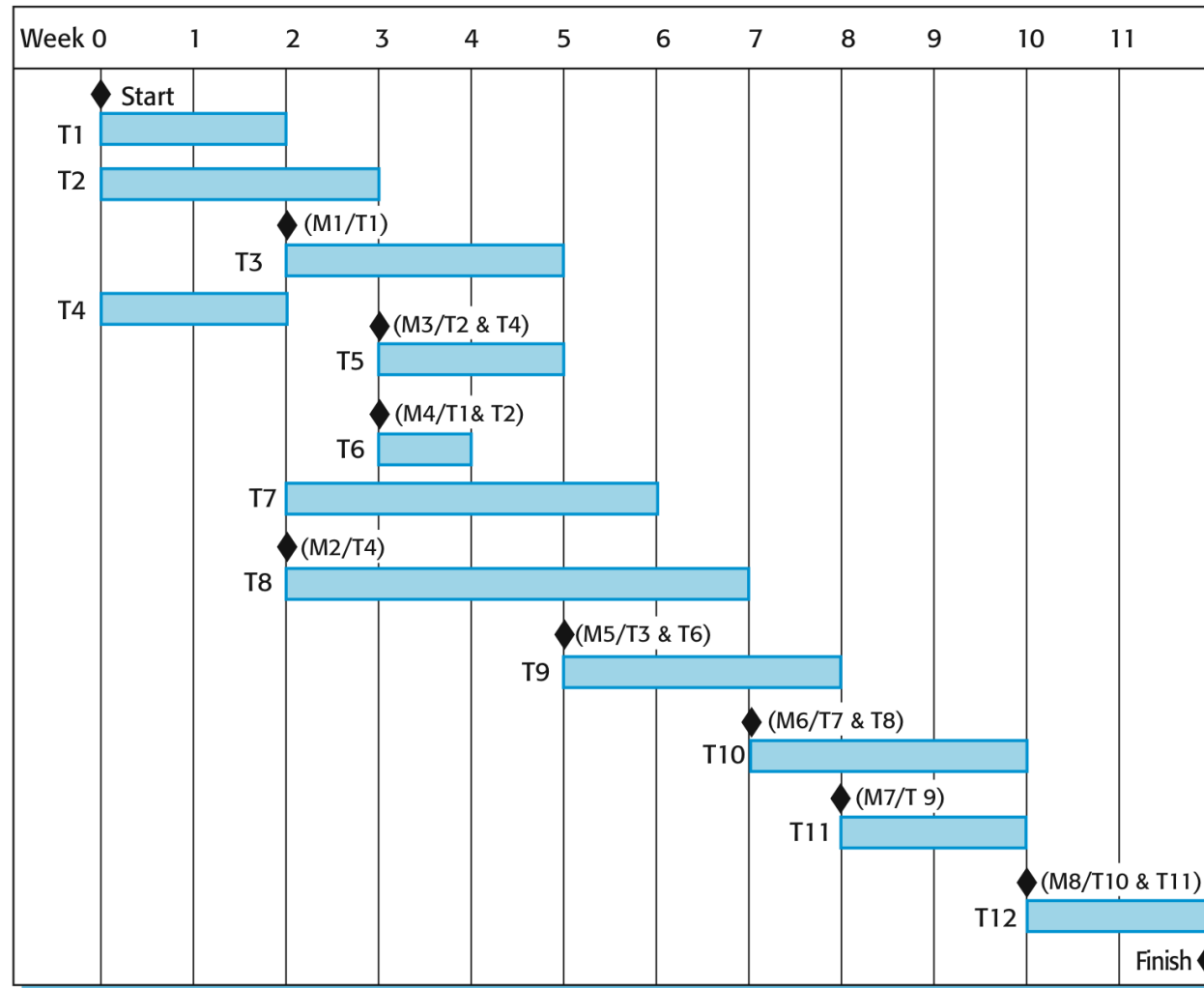
Schedule representation

- ✧ Graphical notations (PERT chart) are normally used to illustrate the project schedule. These show the project breakdown into tasks. Tasks should not be too small. They should take about a week or two.
- ✧ Bar charts (Gantt chart) are the most commonly used representation for project schedules. They show the schedule as activities or resources against time.

Graphical notations (PERT chart)



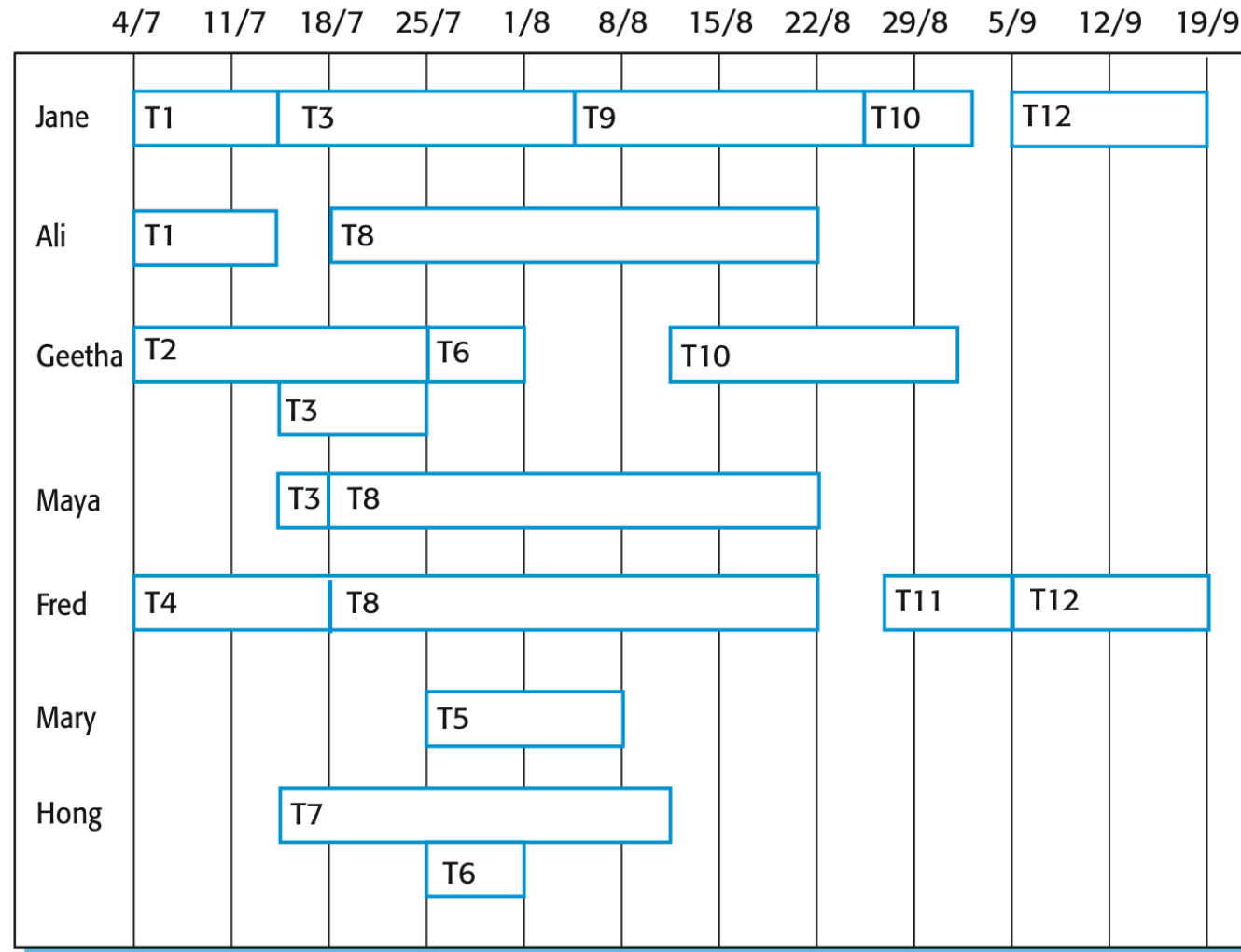
Activity bar chart (Gantt chart)



Tasks,
durations, and
dependencies

Task	Duration (days)	Dependencies
T1	10	
T2	15	
T3	15	T1 (M1)
T4	10	
T5	10	T2, T4 (M3)
T6	5	T1, T2 (M4)
T7	20	T1 (M1)
T8	25	T4 (M2)
T9	15	T3, T6 (M5)
T10	15	T7, T8 (M6)
T11	10	T9 (M7)
T12	10	T10, T11 (M8)

Staff allocation chart



Scheduling problems

- ✧ Estimating the **difficulty of problems** and hence the cost of developing a solution is hard.
- ✧ **Productivity** is not proportional to the number of people working on a task.
- ✧ **Adding people** to a late project makes it later because of communication overheads.
- ✧ **The unexpected** always happens. Always allow contingency in planning.

Risk Management

- "risk" is a **problem that could cause some loss or threaten the progress of the project, but which has not happened yet.**
- **Risk analysis and management** is the process of identifying, addressing, and eliminating potential problems before they have a negative impact on the project.

Risk Management

- **Risk Identification:** entails brainstorming. It also entails the creation of a risk list. The preparation of a risk list entails identifying risks that have occurred repeatedly in previous software projects.
- **Risk Assessment and Prioritization:** It is a procedure within project management that includes the following steps:
 - Identifying the issues that are causing risk in projects
 - Determining the likelihood of a problem occurring
 - Determining the problem's impact
 - Assigning probability and impact values ranging from 1 to 10

Risk Response Strategies

- **Risk Avoidance and Mitigation:** The goal of this technique is to eliminate the occurrence of risks entirely.
 - To avoid risks, reduce the scope of projects by eliminating non-essential requirements.
 - Risk avoidance involves identifying potential risks and then eliminating them as much as possible, or reducing their impact if they cannot be eliminated.
- **Risk transfer:** This technique is used in software engineering to reduce the risk of a project. Risk transfer is usually used when the scope of a project is too large for any one team to handle, and there is no way to split up the work so that each team can be responsible for its own piece of it. In this case, you have to find an outside company that can take on some portion of your project.

Risk Response Strategies(cont'd)

- **Risk acceptance:** is a technique that involves taking on risks in order to complete the system. In this case, it makes sense to accept some level of risk in order to have time to figure out what needs to be done and how long it will take. The only way to know whether this will work is by trying it out, you may find that you were right all along, or you may discover that you need more time than expected.
- **Risk Monitoring:** The risk should be continuously monitored, re-evaluating the risks, the impact of the risk, and the probability of the risk occurring. This guarantees that: The dangers have been discovered and reduced. The magnitude and impact of risk are assessed.

Kinds of Risks in Software Development

1.Schedule Risk : Schedule related risks refers to time. The wrong schedule affects the project development and delivery.

Some reasons for Schedule risks :

1. Time is not estimated perfectly
2. Improper resource allocation
3. Frequent project scope expansion
4. Failure in function identification and its' completion

Kinds of Risks in Software Development

2. Budget Risk :

Budget related risks refers to the monetary risks mainly it occurs due to budget overruns.

Some reasons for Budget risks :

1. Wrong/Improper budget estimation
2. Unexpected Project Scope expansion
3. Mismanagement in budget handling
4. Cost overruns

Kinds of Risks in Software Development

3. Operational Risks :

Operational risks which happen in day-to-day operational activities during project development due to improper process implementation or some external operational risks.

Some reasons for Operational risks :

1. Insufficient resources
2. Conflict between tasks and employees
3. Less number of skilled people
4. Lack of communication and cooperation
5. Lack of clarity in roles and responsibilities
6. Insufficient training.

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Kinds of Risks in Software Development

4. Technical Risks :

Technical risks refers to the functional risk or performance risk which means this technical risk mainly associated with functionality of product or performance part of the software product.

Some reasons for Technical risks :

1. Frequent changes in requirement
2. Less use of future technologies
3. Less number of skilled employee
4. High complexity in implementation
5. Improper integration of modules

Kinds of Risks in Software Development

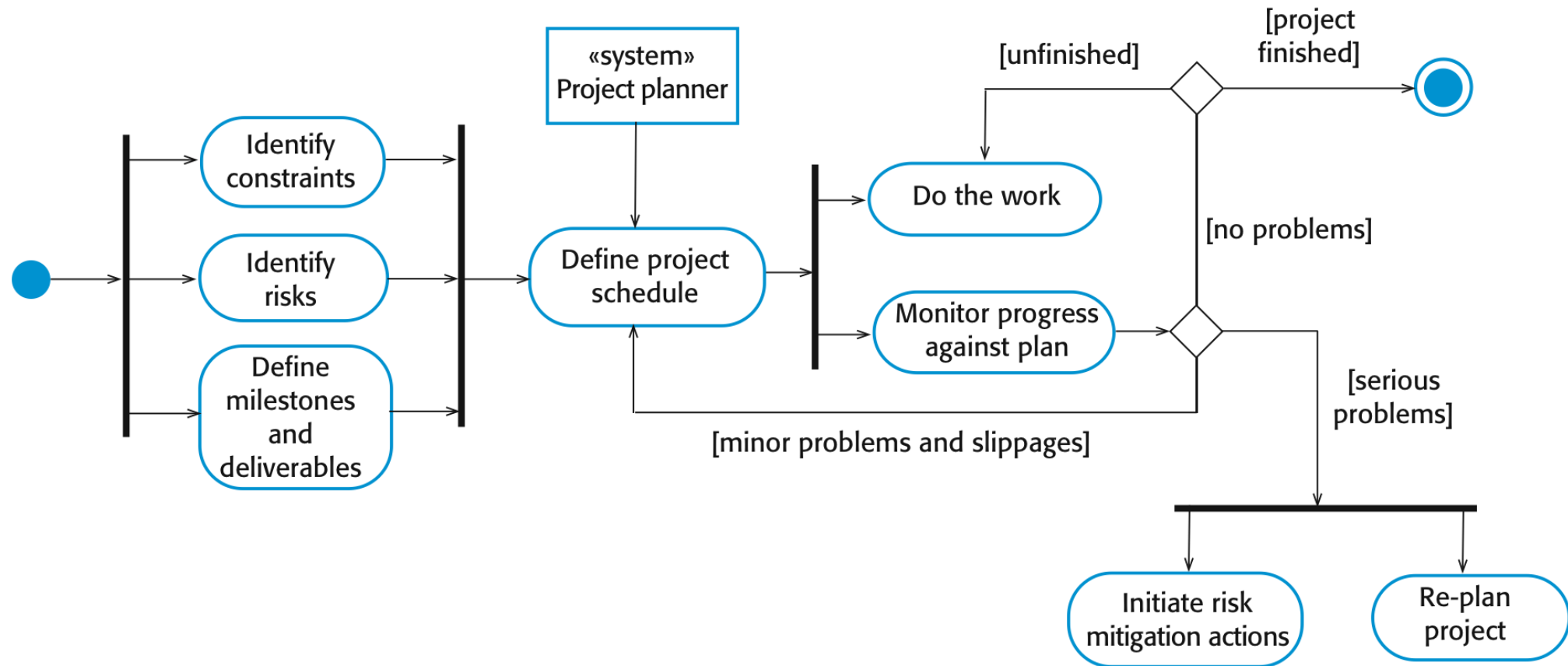
5. Programmatic Risks :

Programmatic risks refers to the external risk or other unavoidable risks. These risks come from outside and it is out of control of programs.

Some reasons for Programmatic risks :

1. Rapid development of market
2. Running out of fund / Limited fund for project development
3. Changes in Government rules/policy
4. Loss of contracts due to any reason

The project planning process



THE END

- <https://www.youtube.com/watch?v=i-QyW8D3ei0>